

- 10.1 a) For a CMOS circuit using flip-flops, the maximum logic propagation delay t_{pd} is defined by the inequality

$$t_{pd} \leq T_c - (t_{\text{setup}} + t_{pcq}),$$

where T_c is the clock period, t_{setup} is the setup time of the device(s), and t_{pcq} is the clk-to- Q delay of the device(s). For a single flip-flop, using the values given in Table 10.5, the maximum logic propagation delay t_{pd} , with all values given in picoseconds, may be calculated as

$$t_{pd_{\text{max}}} = T_c - (t_{\text{setup}} + t_{pcq}) = 500 - (65 + 50) = 500 - 115 = 385 \text{ ps.}$$

- b) For a CMOS circuit consisting of a two-phase latch (two series connected transparent latches) with no overlap time t_{overlap} , the total maximum logic delay is the sum of the individual maximum logic delays of each latch, which, similar to the above case, means the total maximum logic delay may be defined by the constraint

$$t_{pd} \leq T_c - (2t_{pdq}),$$

so that the maximum allowable logic delay may be calculated, using given latch parameters from Table 10.5, with all values given in picoseconds, as

$$t_{pd_{\text{max}}} = T_c - 2t_{pdq} = 500 - 2(40) = 500 - 80 = 420 \text{ ps.}$$

- c) Solving similarly for the maximum logic delay for two pulsed latches, since the pulse is wide enough to hide the setup time (80 ps > 25 ps, given in picoseconds and referencing Table 10.5, is given by

$$t_{pd_{\text{max}}} = T_c - \max(t_{pdq}, t_{pcq} + t_{\text{setup}} - t_{pw}) = 500 - 40 = 460 \text{ ps.}$$

- 10.2 Accounting for clock skew of up to 50ps in all cases, and using a similar methodology to the above, with all values given in picoseconds

- a) For a single flip-flop,

$$t_{pd_{\text{max}}} = T_c - (t_{pcq} + t_{\text{setup}} + t_{\text{skew}}) = 500 - (50 + 65 + 50) = 500 - 165 = 335 \text{ ps.}$$

- b) For two transparent latches, clock skew has no impact on performance, and so

$$t_{pd_{\text{max}}} = 2t_{pdq} = 420 \text{ ps.}$$

- c) For pulsed latches, though they are similarly skew-tolerant, this only applies if the pulse is wide enough. In this case, t_{pdq} is less than the total sequencing overhead, the latches are not *totally* skew-tolerant, and so

$$t_{pd_{\text{max}}} = T_c - (t_{pcq} + t_{\text{setup}} - t_{pw} + t_{\text{skew}}) = 500 - (50 + 25 - 80 + 50) = 500 - 45 = 455 \text{ ps.}$$

- 10.3 a) For a single flip-flop, the minimum contamination delay may be derived, with all values given in picoseconds, as

$$t_{cd_{\text{min}}} = t_{\text{hold}} - t_{ccq} = 30 - 35 = -5 \text{ ps}$$

- b) For two phase-transparent latches with 50% duty cycle clocks,

$$t_{cd_{\text{min}}} = t_{\text{hold}} - t_{ccq} = -5 \text{ ps,}$$

for each latch.

- c) For the same configuration with 60 ps of no overlap between phases,

$$t_{cd_{\text{min}}} = t_{\text{hold}} - t_{ccq} - t_{\text{nooverlap}} = 30 - 35 - 60 = -70 \text{ ps.}$$

- d) Finally, for pulsed latches with an 80 ps pulse width,

$$t_{cd_{\text{min}}} = t_{\text{hold}} - t_{ccq} + t_{pw} = 30 - 35 + 80 = 75 \text{ ps.}$$

10.4 Incorporating a maximum clock skew of 50 ps,

a)

$$t_{cd_{\min}} = t_{\text{hold}} - t_{ccq} + t_{\text{skew}} = 30 - 35 + 50 = 45\text{ps}$$

b)

$$t_{cd_{\min}} = t_{\text{hold}} - t_{ccq} + t_{\text{skew}} = 30 - 35 + 50 = 45 \text{ ps},$$

for each latch.

c)

$$t_{cd_{\min}} = t_{\text{hold}} - t_{ccq} = 30 - 35 = -5 \text{ ps},$$

d)

$$t_{cd_{\min}} = t_{\text{hold}} - t_{ccq} + t_{pw} + t_{\text{skew}} = 30 - 35 + 80 + 50 = 95 \text{ ps}.$$

10.4 a) For a cycle time of $T_C = 1200$ ps at a 50% duty cycle,

$\Delta 1 : 550 < 600$ no borrowing

$\Delta 2 : 580 < 600$ no borrowing

$\Delta 3 : 450 < 600$ no borrowing

$\Delta 4 : 200 < 600$ no borrowing

For a cycle time of $T_C = 1000$ ps at a 50% duty cycle,

$\Delta 1 : 550 < 500$ borrowing 50ps

$\Delta 2 : 580 > 450$ borrowing 130ps

$\Delta 3 : 450 > 370$ borrowing 80ps

$\Delta 4 : 200 < 420$ no borrowing

For a cycle time of $T_C = 800$ ps at a 50% duty cycle,

$\Delta 1 : 550 > 400$ borrowing 50ps

$\Delta 2 : 580 < 350$ borrowing 330ps

$\Delta 3 : 450 < 70$ violated time requirement

$\Delta 4 : 200 > 400$ does not execute